Manifold for Type 1 Diabetes – Restoration

Network for Pancreatic Organ Donors with Diabetes (nPOD) — tissue-level causal integration across T1D disease stages

1 · THE CLINICAL PROBLEM

nPOD occupies a position in T1D research that no other resource can match: the collection and characterization of pancreatic tissue from organ donors across the full disease spectrum — at-risk autoantibody-positive, new-onset, established, and long-duration T1D — paired with recipient clinical history, multi-omic tissue characterization (histology, single-cell transcriptomics, spatial transcriptomics, proteomics), and a distributed investigator network. The resource has substantially redefined T1D pathology in the last fifteen years. The analytic frontier ahead of the current methodology is structural:

- **Tissue-level causal inference across the disease spectrum.** Features of pancreatic tissue (islet mass, insulitis intensity, β-cell senescence markers, endocrine-exocrine coupling, residual β-cell phenotype) vary systematically across nPOD disease-stage strata. Recovering the directed structure — which tissue-level features are consequence, which are cause, which are upstream drivers of progression — is a causal-inference problem that parallel-correlation analyses do not address.
- **Multi-omic integration across modalities.** Single-cell transcriptomics, spatial transcriptomics, proteomics, and clinical-history covariates cohere biologically but are typically analyzed within-modality. Cross-modality directed-graph inference is the tool the integrated data volume warrants.
- **Distributed-investigator analytic coherence.** nPOD-affiliated investigator groups produce analyses on overlapping tissue panels. A shared **Ω**F decomposition provides a standardized comparative layer across investigator analyses — a methodological contribution the consortium has separately identified as a field-level need.

Three capabilities remain underdeveloped in current nPOD-linked analytic practice:

- Unified causal modeling across pancreatic-tissue disease-stage strata — recovering the directed structure connecting autoantibody profile, islet mass, insulitis, β-cell phenotypic signatures, and residual function, rather than stratum-by-stratum characterization.
- Cross-modality integration of histological, transcriptomic, proteomic, and clinical-history data into a single causal substrate.
- Composite-fragility decomposition (ΩF) as a standardized comparative layer across nPOD-affiliated investigator analyses — enabling cross-study benchmarking within the consortium on a common analytic footing.

> nPOD is the only resource in T1D research producing pancreatic tissue across the full disease spectrum paired with deep clinical history and multi-omic characterization. The outstanding question — what drives progression from at-risk autoantibody status to clinical disease to established hypoinsulinemia at the tissue level — is no longer a descriptive problem but a causal-inference problem on the integrated nPOD substrate.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities form the basis of the nPOD-configured deployment:

1. **Causal discovery (Spirtes / PC algorithm) on nPOD multi-modal tissue data.** Recovers the directed structure among histological features, transcriptomic and proteomic signatures, autoantibody profile, and disease-stage clinical covariates, providing the substrate for stage-transition causal inference.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Configured for tissue-level disease-stage data — persistent homology on multi-omic tissue manifolds, heterogeneous-exposure-effect learners for autoantibody profile and environmental-history covariates, Bayesian change-point on disease-stage transition markers — with the composite score as a standardized layer for cross-investigator comparison.
3. **Pearl do-calculus and interdiction on the tissue-level graph.** Target prioritization for preservation and regeneration programs is framed as maximum-attenuation intervention on the inferred progression cascade at the tissue level, identifying the tissue-level pathway junctions where interventions produce maximum downstream attenuation.

3 · ΩF PILLAR MAPPING – PANCREATIC TISSUE PHYSIOLOGY

ΩF PILLAR	NPOD TISSUE READING
I — Interaction	Autoantibody × HLA × tissue-feature interactions; immune-cell infiltrate × residual β-cell phenotype × endocrine-exocrine coupling
R — Reserve	Residual islet mass by histology; β-cell functional-marker density; residual insulin / C-peptide immunoreactivity across disease-stage strata
J — Junction	Insulitis intensity and immune-cell infiltrate composition at the islet boundary; endocrine-exocrine interface signatures
C — Cascade	β-cell senescence and dedifferentiation cascades in situ; ER-stress and apoptosis markers; epitope-spreading signatures across disease-stage strata
T — Temporal	Disease-stage progression hazard; time-to-established-T1D from autoantibody-positive state given tissue-level and clinical covariates

4 · CRITICALITY ESTIMATOR CONFIGURATION

nPOD data is tissue-level, multi-modal, and stratified across disease stages rather than longitudinal per subject. The criticality layer is configured accordingly:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Persistent homology on multi-omic tissue manifolds	Topological regime detection across disease-stage strata; identification of discontinuous phenotypic clusters within-stratum	I, C
Heterogeneous-exposure-effect learners (Causal Forest, BART) on autoantibody and environmental-history profile	Tissue-feature differentiation across autoantibody-profile and environmental-exposure subgroups	I, C
Bayesian change-point across disease-stage strata	Stage-transition boundary detection in tissue-feature space; progression-marker inflection identification	C, T
Spatial-transcriptomic coupling models (Gaussian random field, neighborhood-enrichment)	Insulitis × residual β-cell spatial coupling; endocrine-exocrine interface feature characterization	J, C

Additional estimators available within the same layer for subsequent phases:

- **Transfer entropy** across cross-modality tissue-feature panels — quantifies directional coupling between histological, transcriptomic, and proteomic signatures of β-cell stress and regeneration.
- **Graph neural networks** on the discovered tissue-level causal graph — provides tractable inference under interventional queries for target-prioritization analyses.

5 · PROPOSED INITIAL ENGAGEMENT (PHASE 1)

The scope of the proposed Phase 1 engagement is defined to produce a deliverable of demonstrable analytic value to nPOD and to the broader nPOD-affiliated investigator network within a contained time and data footprint.

- **Primary dataset.** nPOD pancreatic-tissue multi-omic data across at least three disease-stage strata (autoantibody-positive at-risk, new-onset T1D, established T1D), paired with donor clinical history and autoantibody profile. Scope prioritized to modalities already shared across the nPOD investigator network (histology, scRNA-seq, proteomics) to maximize consortium utility of the deliverable.
- **Access path.** nPOD Data Access Committee under the sponsorship of a UF Diabetes Institute principal investigator.
- **Deliverable.** Cross-modality tissue-level causal graph spanning the selected disease-stage strata. Per-donor **Ω**F decomposition across the full five-pillar framework, positioned as a standardized cross-investigator comparative layer. Pearl-do-calculus ranking of tissue-level pathway junctions where preservation and regeneration interventions are predicted to produce maximum downstream attenuation.
- **Success criteria.** Tissue-level causal graph validated against pre-specified biological priors from nPOD and UF investigators; identification of at least one previously unrecognized tissue-level progression pathway; methodological deliverable suitable for publication in an nPOD-affiliated venue and usable by the broader investigator network as a standardized comparative substrate.
- **Timeline.** Fourteen to sixteen weeks from data access, reflecting the cross-modality and cross-stratum integration complexity.
- **Excluded from Phase 1** (reserved for the Phase 2 expansion): integration across all available nPOD modalities simultaneously, prospective tissue-acquisition coordination, bespoke assay development.

6 · RATIONALE AND TIMING

- **nPOD is a uniquely positioned resource.** Pancreatic tissue across the full T1D disease spectrum, paired with deep clinical history and cross-modality multi-omic characterization, does not exist elsewhere. The analytic tools appropriate to tissue-level causal integration at this scale — causal-discovery engines, persistent homology, spatial-transcriptomic modeling — have only recently matured to production grade against it.
- **Preservation and regeneration programs are tissue-driven.** Teplizumab, verapamil, DYRK1A inhibitors, and forthcoming agents operate on tissue-level pathway junctions that nPOD data characterizes better than any patient-trajectory dataset. Principled target prioritization routes through tissue-level causal inference, which is the Manifold deployment's native substrate.
- **Replacement paradigm broadens, not narrows, the tissue-biology question.** Insulin-independence in early-phase Vertex VX-880 cohorts and reproducible human β-cell proliferation via DYRK1A inhibition both reinforce the field's need for tissue-level understanding of what distinguishes residual-function phenotypes — the question nPOD is structured to answer.
- **Consortium-utility framing aligns with funding priorities.** nPOD, Breakthrough T1D, Helmsley Charitable Trust, and NIH NIDDK have each prioritized methodological deliverables that serve the broader investigator network; a standardized **Ω**F decomposition is structured to meet that requirement.
- **The platform is production-ready.** Manifold's causal, criticality, and interdiction engines are production-grade; configuration for nPOD tissue-level multi-modal data represents bounded, well-scoped work.

7 · PROPOSED PARTNERSHIP

Partnership with a University of Florida Diabetes Institute principal investigator affiliated with nPOD is sought for:

1. Retrospective nPOD cross-modality tissue data access under the nPOD Data Access Committee framework.
2. Co-authorship on the tissue-level causal-integration methodological manuscript, positioned as an nPOD-affiliated consortium deliverable.
3. Co-application to NIH NIDDK, Breakthrough T1D, or the Helmsley Charitable Trust for the Phase 2 arm extending integration across additional modalities and disease-stage strata.

In exchange, Apex Analytica contributes the causal-inference engine, the **Ω**F framework, and the engineering team. Apex retains commercial rights to the platform; research intellectual property from the joint work is shared on standard academic–industry terms.